Computer Networks
Fall 2024/25

Exercise-3

Submission in pairs by Thursday, **26-12-2024.** Submission is done by uploading your work to the course Moodle website. >>> No late submissions will be accepted
The name of the submitted file must be Exercise3_username1_username2**.pdf**
For example: Exercise3_israel.israeli_david.davidi.zip (with dots). The first and last names of both students must appear. Only one student should submit the assignment (not both). It Covers Chapter 3 from the textbook. Please notice that once the grades for an exercise are published, appeals can only be submitted in the following five days. The grader is Mr. Daniel Ohayon, and you may contact him by email at *reichman.computer.networks@gmail.com*.

Problem 1 (20 points).

1. The question refers to the Go-Back-N protocol. The sender's code (from Lecture 6, slide 45) has been changed so that it is possible to send only **one** packet when a timeout happens. When there is a timeout, only the first packet (the packet that the send_base pointer is pointing at) is resent. The code of the receiver has not been changed. Is the protocol correct after the change? If yes, justify and explain how it differs from the behavior of the original algorithm; if no, give an example of an incorrect behavior.

2. Consider the *sliding window* protocol with a sender window size **15** and sequence number range from 1 to 532. Suppose that at time ***t***, the last packet in the sender's window has a sequence number **150**. Assume that the medium does not reorder messages in both directions. What are the ACK messages with different sequence numbers that can be currently propagating back from the receiver to the sender at time *t*
and what are their number assuming that:

1. The GO-BACK-N protocol is used.
2. The Selective Repeat protocol is used.

Problem 2 (20 points).

To get around the problem of sequence numbers wrapping around while old packets still exist, one could use **58**-bit sequence numbers. However, theoretically, an optical fiber can run (i.e., transmit) at 7.5 Tbps (**7.5** x 10^{12} **bits** per second). Assume that each packet is of size **1500 bytes** and has a lifetime after which it disappears if it does not arrive at its destination. Assume that each byte has its own sequence number, as TCP does. What is the biggest packet lifetime requested to ensure the networks do not have a wraparound problem?
Notice that the optical fiber transmission rate mentioned above is in **bits, not bytes**.
Try that the length of your answer will be no more than six lines.

Problem 3 (20 points).

Host A and host B are communicating over a TCP connection; the first segment sent from A to B has sequence number 1, and B has already received from A all (data) bytes up through byte **6000**. Suppose A sends five new segments, S1, S2, S3, S4, and S5, to B back-to-back. S1 is sent first, S2 second, S3 third, S4 fourth, and S5 fifth. S1, S2, S3, S4, and S5 contain 100, 200, 300, 400, and 500 bytes of data, respectively. Assume that delayed-ack is not used (an ACK is sent as soon as a packet arrives).

1. If S2 arrive s at B before S4 and S5 and after S1 and S3, what is the acknowledgment number sent by B immediately after S2 arrives? Assume that the S3 arrives before S1 and S4 arrives before S5.

2. If S1 arrives at B after S2 and S3, and before S4 and S5, what is the acknowledgment number sent by B immediately after S1 arrives? Assume that the S3 arrives before S2 and S5 arrives before S4.

3. If S3 arrives at B before S4, and after S1, S2 and S5, what is the acknowledgment number sent by B immediately after S3 arrives? Assume that the S2 arrives before S1 and after S5.

4. If S4 arrives at B after S1, S2, and S3 and before S5, what is the acknowledgment number sent by B immediately after S4 arrives? Assume that S3 arrives before S2 and after S1.

5. If S5 arrives at B after S1, S2, S3, and S4, what is the acknowledgment number sent by B immediately after S4 arrives? Assume that the S2 arrives before S3 and after S1 and that S4 arrives before S3 and after S2.

For each of the questions above, give an answer assuming (1) buffering is not used on the receiver side and (2) buffering is used on the receiver side.

Answer:

#	**Without buffering**	**With buffering**
1		
2		
3		
4		
5		

Briefly justify your answers.

Problem 4 (20 points).
Twelve packets are sent over TCP, with a **1**ms difference in the submission time. The first packet is sent at time **0**. The receivers send ACK immediately when receiving a packet. The RTT is **8**ms (half RTT is 4ms), and the timeout time is **10**ms. The transmission time (that is, putting the data on the line) for the packets and ACK is negligible. Assume that the TCP window size is always big enough, so there is no need to wait to send new packets. Also, assume that all packets sent before time 0 were already acknowledged. There is no use of "delayed Ack" (an ACK is sent as soon as a packet arrives). A packet that is sent again can be sent immediately.

1. (3 points) Assume that the ACK for the seventh packet (sent at time 6) was lost. Also, assume that the fast retransmit is used and no more packets are lost. When will the sender know that the seventh packet has arrived?

2. (3 points) Assume the seventh packet (sent at time 6) was lost. Also assume that the fast retransmit is not used, and no more packets are lost. When will the seventh packet arrive (to the receiver)?

3. (3 points) Assume the third packet (sent at time 2) was lost. Also assume that the fast retransmit is used, and no more messages are lost. When will the third packet arrive (to the receiver)?

4. (3 points) Assume the third packet (sent at time 2) was lost. In addition, also the fourth, sixth and seventh packets were lost. Assume that the fast retransmit is used, and no more packets are lost. When will the third packet arrive (to the receiver)?

5. (3 points) Assume the third packet (sent at time 2) was lost. In addition, also the fourth, sixth and seventh packets were lost. Assume that the fast retransmit is not used, and no more packets are lost. When will the third packet arrive (to the receiver)?

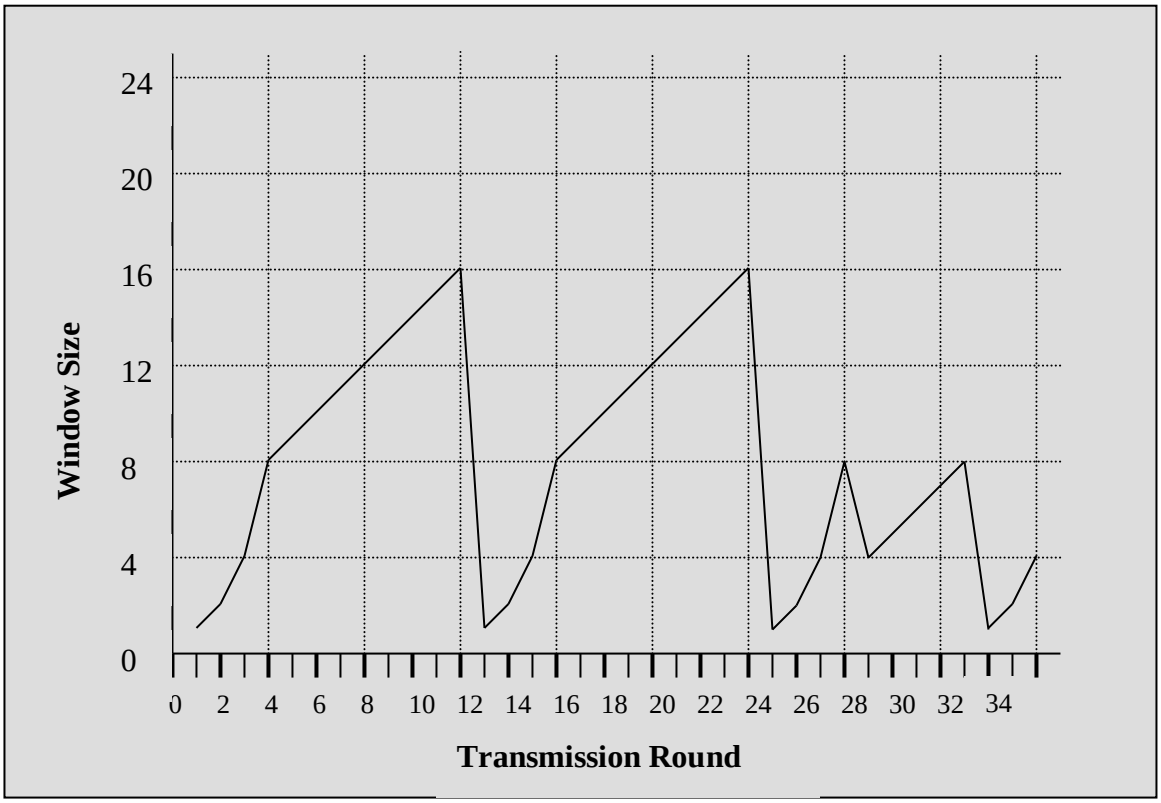
6. (5 points) Assume the second packet (sent at time 1) was lost. In addition, also the third, fourth, fifth, sixth, seventh, eighth and ninth packets were lost. Assume that the fast retransmit is used, and no more packets are lost. When will the third packet arrive (to the receiver)?

question	Answer (in msec)
1	
2	
3	
4	
5	
6	

Briefly justify your answers.

Problem 5 (20 points).

Consider the following plot of TCP window size as a function of time that is, as a function of the number of rounds. (In each round, the sender transmits all the segments in its congestion window and either receives acknowledgments for them or there is a loss event.)



If TCP Reno is the protocol experiencing the above behavior, answer the following questions.

a. Identify all the intervals of time when TCP slow start is operating, and identify the intervals of time when TCP congestion avoidance is operating.

b. What are all the transmission rounds in which segment loss is detected by a triple duplicate ACK?

c. What are all the transmission rounds in which segment loss is detected by a timeout?

d. What are all the values of the Threshold during the first 34 transmission rounds (starting from round 1)?

e. During what transmission round is the 75th segment sent?